

Zhoutian Xu

E: zhoutianxuc@gmail.com | P: (217)-305-0464 | <https://www.linkedin.com/in/zhoutian-xu>

Education

Cornell University

Ithaca, NY

Master of Engineering in Operations Research and Information Engineering

Aug 2024 – Expected Dec 2025

Selected Coursework: Artificial Intelligence, NLP, Deep Learning, Reinforcement Learning, Machine Learning, Big Data

University of Illinois at Urbana-Champaign

Champaign, IL

Bachelor of Science in Industrial Engineering, Minor in Statistics

Aug 2019 – Dec 2023

Selected Coursework: Optimization, Statistical Learning, Algorithm Design, Data Analysis

Graduated with Honors

- Bernt O. Larson Project Design Award, Second Place

Skills

Technical: C++, Python, R, SQL, MATLAB, AMPL, Pytorch, Gurobi, CPLEX, Arena Simulation, Minitab, Tableau, Power BI

Professional Experience

Wegmans (Capstone)

Rochester, NY

Data Science Consultant

Dec 2024 – Present

- Develop a product claim prediction model by integrating order data with external factors using **XGBoost** and **LSTM** to precisely identify top high-risk orders, supporting supplier management and pre-inspection strategy.

Tesla

Shanghai, China

Data Scientist Intern (Industrial AI)

Apr 2024 – Aug 2024

- Developed an **Anomaly Detection Algorithm** using **Python** for welding processes, encapsulated the algorithm using **FastAPI**, developed an **ETL** pipeline to process over **700k** rows of welding data per day, and visualized processed data as a real-time 3-D scatter plot, reducing reaction time by **90%** and saving over **\$75,000** per year.
- Implemented **DBSCAN** and **K-means** clusterings on battery electrode data, enhancing **LightGBM** model performance and revealing causations for **78.5%** of product defects.
- Built a **Fast Dynamic Time Warping** model to identify the pseudo-soldering defect and uncover its generalized resistance patterns by pinpointing products with the highest total DTW distances across batches.
- Contributed to **Pricing Strategy Optimization** for dynamic electricity pricing system of EV charging stations (Superchargers) through customer profile analysis, pricing elasticity modeling, and future demand forecasting.

The Walt Disney Company

Shanghai, China

Industrial Engineering Intern

Jun 2023 – Aug 2023

- Led a study to provide data-driven insights highlighting the expansion requirements of two backstage canteens to meet the rising peak hour demands and long-term growth in staff, and developed expansion and layout plans aligned with anticipated demand and growth trends.
- Discovered bottlenecks in guest serving across in-resort activities through **Regression Analysis** and strategically streamlined procedures, contributing to a **~15%** reduction in customer waiting time.
- Designed a **Queue Optimization** model with the **Gurobi** library in **Python**, and validated its effectiveness through a **Stochastic Simulation** model incorporating guest behaviors and randomness, contributing to reduced waiting friction and a **10%** improvement in ride vehicle utilization.

Fujian Benz Auto Co., Ltd (R&D Center of the Mercedes-Benz Group)

Fuzhou, China

R&D Engineer Intern

May 2023 – Jun 2023

- Developed an **Automated Data Processing Script** in **Python** to efficiently process over **100k** rows of mechanical production-related data per day, enabling effortless elimination of extraneous data noise and tailored data integration, outperforming existing VBA script in processing speed by **70%**.
- Built real-time global **Data-sharing Dashboards** using **Power BI**, empowering global manufacturing centers to grasp production and maintenance progress and monitor order dynamics promptly.

Research Experience

Reinforcement Learning-based Quantization & Pruning for LLM

Ithaca, NY

Graduate Student Researcher (Supervisor: Dr. Xi Cheng)

Jan 2025 – Present

- Develop an algorithm-driven **mixed-precision quantization** framework utilizing hierarchical reward functions to achieve **model compression** and enhanced inference speeds.
- Design a sensitivity analysis-based tiered quantization scheme for KV Cache, implementing hybrid storage strategies for attention head key-value vectors.